Jinbin Luo

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Education

Shanghai Jiao Tong University Shanghai, China
B.Eng. in Computer Science & Technology Sep. 2023 – Jun. 2027 (expected)

- Member of the ACM Honor Class, a research-oriented CS program for students ranked in the top 2% at SJTU.
- **Core GPA** 3.96/4.30 **Rank** 5/27
- **Selected core courses** Compiler Design and Implementation (98/100), Operating System (95/100), Computer Architecture (90/100), Principle and Practice of Computer Algorithms (100/100), Machine Learning (91/100), Math Analysis (94/100), Linear Algebra (95/100).

Research Interests

My research interests lie in optimizing large language model (LLM) inference, with a focus on improving efficiency, scalability, and system-level performance.

Experience

Massachusetts Institute of Technology, Han Lab Cambridge, MA
Research Intern, advised by Prof. Song Han Jul. 2026 – Sep. 2026

Shanghai Jiao Tong University, EPCC Lab Shanghai, China
Research Intern, advised by Prof. Quan Chen Jul. 2025 – Jul. 2026

Publications

DynaFlow: Transparent and Flexible Intra-Device Parallelism via Programmable Operator Scheduling
Yi Pan, Yile Gu, **Jinbin Luo**, Yibo Wu, Ziren Wang, Hongtao Zhang, Ziyi Xu, Shengkai Lin, Baris Kasikci, and Stephanie Wang.
Proceedings of Machine Learning and Systems (MLSys), 2026.

Manuscripts Under Review

CacheTide: Transparent Cache Management and Layer-Wise Streaming for Hybrid LLM Serving
Jinbin Luo, Xiaozhe Yao, Shang Yang, Zhijian Liu, Song Han, and Qinghao Hu.
Under review at MLSys 2027.

MoS: Seamless Hybrid Scaling for Cost-Efficient MoE Inference
Junxiao Deng*, **Jinbin Luo***, Weihao Cui, Han Zhao, Siyuan Feng, Ao Xu, Quan Chen, and Minyi Guo.
Under review at MLSys 2027.

* Equal contribution (co-first authors).

Personal Projects

ACore OS Jun. 2025

A Unix-like microkernel for RV64GC. Operating System course project.

- Implemented system calls, user libraries, and a shell.
- Implemented a basic file system.

LLVM JIT Compiler Apr. 2025

A just-in-time compiler for LLVM IR. Advanced Compiler course project.

- Developed a lightweight JIT virtual machine that executes LLVM IR on RISC-V 64.
- Implemented IR parsing, interpretation, and dynamic compilation of hot basic blocks.

Thunder CPU Jan. 2025

An RV32IC processor. Computer Architecture course project.

- Implemented a Verilog CPU using Tomasulo's algorithm for out-of-order execution, with support for the compressed instruction extension.
- Integrated an instruction cache and a branch predictor.

Compiler for the Mx* Language Oct. 2024

A compiler from Mx*, a C++-like language, to RV32I assembly.

- Implemented the compiler in Rust and assembly, using Pest for parsing and LLVM IR as the intermediate representation.
- Implemented conditional constant propagation, aggressive dead code elimination, inlining, mem2reg, and register allocation.

Teaching Experience

- **Principle and Practice of Computer Algorithms** Teaching Assistant for Prof. Yong Yu
- **Data Structure** Teaching Assistant for Prof. Huiyu Weng
- **Computer Programming** Teaching Assistant for Prof. Huiyu Weng

Awards

- Zhiyuan Honorary Scholarship (top 2% at SJTU), 2023 and 2024
- Academic Excellence Scholarship (Second Prize), 2024

Technical Skills

Programming Python, C++, CUDA, Rust, Java, Go, Verilog
Tools Git, CMake, Make, Docker, LaTeX, Markdown
Languages Mandarin (native), English (fluent)